

Program 9

Title: Organize Kubernetes Pods using Labels and Namespaces

Objective: Create isolated environments using namespaces and manage pods efficiently using labels.

Project Structure

```
Program9/
|
├── namespaces.yaml
└── pods.yaml
```

Configuration Files

namespaces.yaml

```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
```

```
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Execution Steps

Step 1: Create Namespaces

```
1rv24mc078_prateek_kumar@arch kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
```

Step 2: Create Pods

```
1rv24mc078_prateek_kumar@arch kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
```

Step 3: List Pods (Default Namespace)

```
1rv24mc078_prateek_kumar@arch kubectl get pods
No resources found in default namespace.
```

Step 4: List Pods in dev Namespace

```
1rv24mc078_prateek_kumar@arch kubectl get pods -n dev
```

NAME	READY	STATUS	RESTARTS	AGE
frontend-pod	1/1	Running	0	1m

Step 5: Filter Pods by Label

```
1rv24mc078_prateek_kumar@arch kubectl get pods -n dev -l app=frontend
```

NAME	READY	STATUS	RESTARTS	AGE
frontend-pod	1/1	Running	0	1m

Step 6: Verify Production Pods

```
1rv24mc078_prateek_kumar@arch kubectl get pods -n prod
```

NAME	READY	STATUS	RESTARTS	AGE
backend-pod	1/1	Running	0	1m

Output

Pods Organized using Labels and Namespaces:

Result

Namespaces were successfully created to isolate environments. Pods were deployed with appropriate labels and efficiently filtered using label selectors and namespaces.